

sedphot

User Manual

Galaxy in, SED photometry out

Version 0.3.0

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1 What this does

sedphot turns a galaxy into an SED-ready photometry table.

You give it a galaxy name (or a position on the sky) and an output directory. It then does some or all of the following, depending on which command you run:

- Looks the galaxy up in the public catalog archives (GALEX, SDSS, J-PLUS, Pan-STARRS, Legacy Surveys, AllWISE, HST) and records the published flux of the nearest source in every band.
- Downloads survey images (SDSS, Pan-STARRS, Legacy Surveys, CFHT, HST) and measures the galaxy itself in every band through one identical procedure, so the colors are internally consistent.
- Optionally requests SPHEREx spectrophotometry from IRSA.
- Writes a combined SED figure, quality-assurance figures, and a JSON record of exactly how every number was produced.

Fluxes come out in microjansky on the AB system. Everything for one galaxy lands under one directory that you choose.

2 Installing

These instructions assume the machine has nothing set up yet.

2.1 Get Python

sedphot needs **Python 3.11 or newer** and a stack of scientific libraries (NumPy, SciPy, Astropy, Astroquery, Matplotlib, and others).

Use **Miniconda**. Conda installs Python and the scientific libraries as pre-built binaries, which avoids the compiler toolchain that some of these packages otherwise require, and it keeps each project's libraries in a separate environment so one project cannot break another.

Windows

1. Go to <https://www.anaconda.com/download/success> (or <https://docs.conda.io/projects/miniconda/>) and download the **Miniconda Windows 64-bit installer (.exe)**.
2. Run the installer. Accept the defaults.
3. From the Start menu, open **Anaconda Prompt**. Every command in this manual is typed there.

macOS

1. Download the **Miniconda macOS installer** for your chip (Apple silicon = **arm64**, Intel = **x86_64**) from the same page.
2. Run the installer, then open **Terminal**.

Linux

1. Download the Miniconda Linux **x86_64** installer script from the same page.
2. Run it with **bash Miniconda3-latest-Linux-x86_64.sh**, then open a new terminal.

Check that conda is available:

```
conda --version
```

You should see something like **conda 24.9.2**.

2.2 Get the code

The source lives at <https://github.com/chrishopp12/query-photometry>.

With git. On Windows, install Git for Windows from <https://git-scm.com/download/win> first; macOS and Linux usually have git already. Then:

```
git clone https://github.com/chrishopp12/query-photometry.git
```

This creates a folder named `query-photometry` in the current directory. Move into it:

```
cd query-photometry
```

Without git. Open <https://github.com/chrishopp12/query-photometry> in a browser, click the green **Code** button, choose **Download ZIP**, and unzip it. You get a folder named `query-photometry-main`. Move into it:

```
cd query-photometry-main
```

You are in the right folder if it contains a file named `pyproject.toml`.

2.3 Create an environment and install

Conda route (recommended, all platforms).

```
conda create -n sedphot python=3.12
```

Answer `y` when it asks. Then activate it:

```
conda activate sedphot
```

Your prompt now starts with `(sedphot)`. Install the package:

```
pip install -e .
```

That reads `pyproject.toml` and pulls in every dependency.

If any dependency fails to build (rare, but `reproject` is the usual suspect), install the scientific stack from conda-forge first and then install `sedphot` without re-resolving them:

```
conda install -c conda-forge numpy scipy pandas astropy
```

```
conda install -c conda-forge astroquery matplotlib pillow
```

```
conda install -c conda-forge reproject requests defusedxml
```

```
pip install -e . --no-deps
```

pip / venv route (no conda). Only use this if Python 3.11+ is already installed and on your `PATH`.

Windows (Anaconda Prompt, PowerShell, or Command Prompt):

```
python -m venv .venv
```

```
.venv\Scripts\activate
```

macOS / Linux:

```
python3 -m venv .venv
```

```
source .venv/bin/activate
```

Then, on any platform:

```
pip install -e .
```

2.4 What “editable” means

`pip install -e .` installs the package **in place**: Python runs the source files in the folder you cloned, rather than a copy hidden away in the environment. If you `git pull` a newer version, the change takes effect immediately with no reinstall.

2.5 Verify the install

Three checks, in increasing strength.

Is the command there?

```
sedphot --help
```

Correct output begins:

```
usage: sedphot [-h]
               {resolve,catalogs,measure,spherex,sed,overlay,remeasure,run}
               ...
```

Galaxy in, SED photometry out: multi-archive retrieval and measurement.

Is the right version installed?

```
python -c "import sedphot; print(sedphot.__version__)"
```

Correct output:

```
0.3.0
```

Can it reach the network? This resolves a real galaxy name against the CDS Sesame service:

```
sedphot resolve --name "NGC 4889"
```

Correct output:

```
[resolve] 'NGC 4889' -> RA=195.033738, Dec=+27.977025 (Sesame)
RA   = 195.03373750
Dec  = +27.97702500
label = NGC_4889
```

If all three work, the install is good.

Every later session needs the environment activated first:

```
conda activate sedphot
```

3 Before you start

The mental model is small.

You pick a galaxy. Either by name (`--name "NGC 4889"`, resolved through Sesame, then NED, then SIMBAD) or by explicit position (`--ra 195.033738 --dec 27.977025`). Give one or the other, never both.

You pick an output directory. `--out-dir` is required on every command that writes anything. Everything for that galaxy lands under it, inside a subdirectory called `Photometry/`: the tables, the figures, the downloaded images, and the JSON records. One galaxy, one directory.

Files are named after a label. The label defaults to the sanitized name (NGC 4889 becomes `NGC_4889`) or, for a bare position, an IAU-style designation (`J130008.10+275837.3`). Override it with `--label`. Output files are `<label>_catalog.csv`, `<label>_measured.csv`, `<label>_sed.png`, and so on.

Downloads are cached. The first run of a galaxy downloads images and catalog queries; later runs on the same directory reuse them. Re-running a command is cheap.

That is the whole model. A typical directory choice looks like:

```
sedphot run --name "NGC 4889" --out-dir NGC_4889
```

or, if you keep galaxies together:

Windows

```
sedphot run --name "NGC 4889" --out-dir galaxies\NGC_4889
```

macOS / Linux

```
sedphot run --name "NGC 4889" --out-dir galaxies/NGC_4889
```

Path separators are the only difference between platforms in this manual. Everything else is identical.

4 The verbs

`sedphot` has eight commands.

Verb	What it does	What it writes
<code>resolve</code>	Turns a name into coordinates and an output label.	Nothing. Prints only.
<code>catalogs</code>	Asks each catalog archive for its published photometry of the nearest source.	<code><label>_catalog.csv</code> + sidecar, <code>coverage_catalogs.json</code>

Verb	What it does	What it writes
measure	Downloads survey images and measures the galaxy in every band.	<label>_measured.csv + sidecar, coverage_measure.json, cached images, per-band QA figures, QA/growth_curves.png
spherex	Requests a SPHEREx spectrophotometry table from IRSA.	SPHEREx/table_photometry.<tag>.csv + sidecar, SPHEREx/extractions.json
sed	Draws the combined SED from tables already on disk.	<label>_sed.png
overlay	Draws every archive's matched position on the HST color image.	<label>_overlay.png, cached HST composite
remeasure	Re-reports fluxes from a finished measurement at a different aperture.	Nothing, unless you pass --out (a CSV) or --write-qa (figures).
run	catalogs , then measure , then optionally spherex , then sed .	Everything the stages above write.

--out-dir is **required** on **catalogs**, **measure**, **spherex**, **sed**, **overlay**, and **run**. **resolve** takes no output directory because it writes nothing. **remeasure** takes a path to a finished measurement's JSON record instead.

5 Your first run

This walks one galaxy from nothing to a measured table, using only the Legacy Surveys images so it finishes in a couple of minutes.

Step 1. Activate the environment.

```
conda activate sedphot
```

Step 2. Confirm the galaxy resolves.

```
sedphot resolve --name "NGC 4889"
```

```
[resolve] 'NGC 4889' -> RA=195.033738, Dec=+27.977025 (Sesame)
RA   = 195.03373750
Dec  = +27.97702500
label = NGC_4889
```

Step 3. Fetch the catalog photometry.

```
sedphot catalogs --name "NGC 4889" --all --out-dir NGC_4889
```

The console shows one block per archive, then a summary:

```
[resolve] 'NGC 4889' -> RA=195.033738, Dec=+27.977025 (Sesame)
```

```

Target: RA=195.033738, Dec=+27.977025   (search radius=2.0")

=== galex ===
[GALEX GUVcat] Attempt 1/5, radius=2.0"
[GALEX GUVcat] Found 1 match(es) at radius=2.0"

=== sdss ===
[SDSS] Attempt 1/5, radius=2.0"
[SDSS] Found 1 match(es) at radius=2.0"

...

Provider summary:
galex          ok                2 rows
sdss           ok                5 rows
jplus          no_match          0 rows -- no J-PLUS DR3 source found
panstarrs      ok                5 rows
legacy         ok                7 rows
allwise        ok                4 rows
hst            ok                2 rows

Saved 25 photometric points to: NGC_4889/Photometry/NGC_4889_catalog.csv

```

followed by the table itself.

Step 4. Measure the images.

```
sedphot measure --name "NGC 4889" --instruments legacy --out-dir NGC_4889
```

```

[resolve] 'NGC 4889' -> RA=195.033738, Dec=+27.977025   (Sesame)

Target: RA=195.033738, Dec=+27.977025   (aperture 12", scene engine)

=== legacy images ===
3 band image(s) ready

(scene catalog and star queries report here on a first run)

[Legacy_r] seat solve [target:sersic]: nfev 168, 4.1s last, ...
[Legacy_r] background track: -0.0074 -> -0.0071 uJy/as2 (6/121 ...
[Legacy_r] fill: 4% of aperture masked, 100% twin-filled, ...
[Legacy_r] f_ap = 30184.2 +/- 41.6 uJy (empap), excess -22.4, ...
Legacy r: 30184.2 +/- 41.6 uJy (empap; QA Legacy_r.png)
[Legacy_z] ...
Legacy z: 48120.7 +/- 63.9 uJy (empap; QA Legacy_z.png)
[Legacy_g] ...
Legacy g: 12507.5 +/- 27.3 uJy (empap; QA Legacy_g.png)

Provider summary:
legacy         ok                3 rows -- measured bands: r, z, g

```

```
Saved 3 measured bands to: NGC_4889/Photometry/NGC_4889_measured.csv
```

followed by the table. The numbers depend on the galaxy; the structure does not. The indented [Legacy_r] lines are the engine's running commentary and are truncated above to fit the page; on screen they run to the end of the line.

Step 5. Draw the SED.

```
sedphot sed --out-dir NGC_4889
```

```
[sed] wrote NGC_4889/Photometry/NGC_4889_sed.png
```

The resulting file tree.

```
NGC_4889/
+-- Photometry/
    +-- NGC_4889_catalog.csv
    +-- NGC_4889_catalog.provenance.json
    +-- NGC_4889_measured.csv
    +-- NGC_4889_measured.provenance.json
    +-- NGC_4889_sed.png
    +-- coverage_catalogs.json
    +-- coverage_measure.json
    +-- Legacy/
        | +-- legacy_ls-dr9_g.fits
        | +-- legacy_ls-dr9_r.fits
        | +-- legacy_ls-dr9_z.fits
        | +-- QA/
        |     +-- Legacy_g.png
        |     +-- Legacy_r.png
        |     +-- Legacy_z.png
    +-- QA/
        | +-- growth_curves.png
    +-- scene/
        +-- gaia_scene.csv
        +-- tractor_scene_dr9.csv
```

Steps 3, 4 and 5 together are exactly what the `run` verb does:

```
sedphot run --name "NGC 4889" --out-dir NGC_4889
```

6 Common tasks

Each entry gives when to use it and the exact command. Substitute your own galaxy name and output directory.

6.1 A note on long commands

A few commands below are too wide for one line and are shown broken across two, with a trailing `\`. That is the continuation character for `bash` and `zsh` (macOS, Linux).

Shell	Continuation character
macOS / Linux (bash, zsh)	\
Windows Command Prompt / Anaconda Prompt	^
Windows PowerShell	` (backtick)

On Windows, either swap the trailing \ for the right character, or — simplest and always correct — delete the line break and paste the whole thing as one long line.

6.2 Look up a galaxy's coordinates from its name

Use this before anything else, to confirm the name resolves and to see what label the output files will use.

```
sedphot resolve --name "NGC 4889"
```

If the name does not resolve, look the position up yourself (NED or SIMBAD in a browser) and use `--ra` / `--dec` everywhere instead:

```
sedphot resolve --ra 195.033738 --dec 27.977025
```

6.3 Fetch catalog photometry from every archive

The normal starting point. Queries all seven catalog archives and writes one combined table.

```
sedphot catalogs --name "M87" --all --out-dir M87
```

Each archive starts with a 2-arcsecond search radius and doubles it up to five times if nothing is found. Raise the starting radius for a galaxy whose catalog position is offset:

```
sedphot catalogs --name "M87" --all --radius 5 --out-dir M87
```

6.4 Fetch catalog photometry from one archive

Use this when you only need one archive, or when one archive was down during an earlier run and you want to retry just that one. The valid names are `galex`, `sdss`, `jplus`, `panstarrs`, `legacy`, `allwise`, `hst`.

```
sedphot catalogs --name "M87" --instruments legacy --out-dir M87
```

Several at once:

```
sedphot catalogs --name "M87" --instruments legacy sdss --out-dir M87
```

Note that the table is rewritten from scratch each time, so a run with one archive replaces a table that held all seven. Re-run with `--all` to rebuild the full table.

6.5 Measure images from every survey

Downloads images from all five image archives and measures the galaxy in every band. This is the slow command: expect minutes per band on a first run, less afterwards because the images are cached.

```
sedphot measure --name "M87" --all --out-dir M87
```

6.6 Measure images from specific surveys only

The valid names are `sdss`, `panstarrs`, `legacy`, `cfht`, `hst`.

```
sedphot measure --name "M87" --instruments legacy cfht --out-dir M87
```

6.7 Measure a subset of filters

`--bands` applies the same filter list to every selected survey; a survey that does not have one of them simply skips it.

```
sedphot measure --name "M87" --all --bands g r z --out-dir M87
```

The default filter sets are:

Survey	Default filters
SDSS	ugriz
Pan-STARRS	grizy
Legacy Surveys (dr9)	grz
Legacy Surveys (dr10)	griz
CFHT MegaCam	ugriz
HST	every filter in the matched observation group

For HST, filter names are flight-filter designations:

```
sedphot measure --name "M87" --instruments hst \
  --bands F475W F814W --out-dir M87
```

6.8 Change the aperture size

The measurement aperture is a circle centered on the target. The default radius is **12 arcseconds**.

```
sedphot measure --name "M87" --all --aperture 15 --out-dir M87
```

The aperture must fall inside the curve-of-growth radius grid, which by default runs from 2 to 40 arcseconds. To go outside that, move the grid too:

```
sedphot measure --name "M87" --all --aperture 50 \
  --radii 2 10 20 30 40 50 60 --cutout-size 200 --out-dir M87
```

6.9 Re-report an existing measurement at a different aperture

This is the fast path. Every measurement stores the *curve of growth* — the enclosed flux at every radius from 2 to 40 arcseconds — so changing the reported aperture is a lookup, not a re-measurement. It takes milliseconds and never touches the files on disk unless you ask it to.

Point it at the measurement's JSON record:

Windows

```
sedphot remeasure M87\Photometry\M87_measured.provenance.json ^
--mode aperture --aperture 10
```

macOS / Linux

```
sedphot remeasure M87/Photometry/M87_measured.provenance.json \
--mode aperture --aperture 10
```

The result prints as a table. To save it, add `--out`:

```
sedphot remeasure M87/Photometry/M87_measured.provenance.json \
--mode aperture --aperture 10 --out M87_R10.csv
```

Two other useful forms. The total flux of the fitted model:

```
sedphot remeasure M87/Photometry/M87_measured.provenance.json \
--integrated
```

And the fitted model’s flux inside a chosen radius (`--mode sersic` is the default):

```
sedphot remeasure M87/Photometry/M87_measured.provenance.json \
--aperture 20
```

Asking for an aperture **past** the stored grid in `--mode aperture` is different: there is no stored value to look up, so the command rebuilds the scene from the saved fit and integrates. It says so, and it takes minutes rather than milliseconds.

6.10 Measure a small or compact target

The defaults are sized for survey-resolution images of ordinary galaxies. A compact target — typically on an HST mosaic — needs four things moved together, or the measurement will treat the galaxy itself as sky.

Flag	Default	Compact-target value
<code>--aperture</code>	12 arcsec	2 arcsec
<code>--cutout-size</code>	120 arcsec	20 arcsec
<code>--sky-rmin</code>	15 arcsec	4 arcsec
<code>--radii</code>	2 to 40 arcsec	0.5 to 8 arcsec

`--sky-rmin` is the boundary between “target” and “sky”. Sky is measured only outside it, and no pixel inside it is allowed to vote on the background level. At the 15-arcsecond default, a compact HST target would have most of the visit excluded from the sky estimate.

```
sedphot measure --name "M87" --instruments hst --aperture 2 \
--cutout-size 20 --sky-rmin 4 --radii 0.5 1 1.5 2 3 4 6 8 \
--out-dir M87
```

The aperture must lie within the `--radii` grid; the grid’s outer radius must be under half the cutout size.

6.11 Run everything for one galaxy in a single command

Catalogs, then images and measurement, then the SED plot. Every stage is isolated: if one fails, the failure is recorded and the rest still run, and the command exits with a nonzero status.

```
sedphot run --name "NGC 4889" --out-dir NGC_4889
```

Add SPHEREx:

```
sedphot run --name "NGC 4889" --spherex sersic --out-dir NGC_4889
```

Leave out slow or irrelevant archives (`--skip` accepts both catalog and image archive names):

```
sedphot run --name "NGC 4889" --skip hst jplus --out-dir NGC_4889
```

`run` accepts every measurement flag `measure` does — `--aperture`, `--bands`, `--cutout-size`, `--sky-rmin`, `--radii`, and the rest — with one exception: `--dump-arrays` is available only on `measure`.

6.12 Make the SED plot from tables already on disk

Use this after editing a table by hand, or after re-running one stage. It reads whichever of `<label>_catalog.csv` and `<label>_measured.csv` exist and draws them on one axis.

```
sedphot sed --out-dir NGC_4889
```

If the directory holds tables for more than one label, name the one you want:

```
sedphot sed --label NGC_4889 --out-dir NGC_4889
```

6.13 Check that the archives all matched the same object

This is the check the flux columns cannot make: two archives confidently reporting photometry for two different objects. It draws each archive's matched position on the HST color image of the field, so a mismatch is visible.

```
sedphot overlay --out-dir NGC_4889
```

It only works where HST has imaged the field. The first run downloads a **~380 MB** file (the HST detection mosaic, needed only for its coordinate grid) and caches it. If you already have a FITS file on the same grid, point at it and skip the download:

```
sedphot overlay --out-dir NGC_4889 \  
  --wcs-from NGC_4889/Photometry/HST/hst_F814W_sci.fits
```

The file must be on the same pixel grid as the color composite; the dimensions are checked, and a mismatch refuses rather than misplacing every marker.

Adjust the panel sizes (both are half-widths in arcseconds):

```
sedphot overlay --zoom-size 3 --context-size 30 --out-dir NGC_4889
```

6.14 Fetch SPHEREx spectrophotometry

Submits a forced-photometry job to the IRSA SPHEREx tool and writes the raw per-visit table. The job runs on IRSA's servers; the command polls until it finishes (up to an hour by default).

The default source model is an elliptical Sérsic profile whose shape is looked up in the Legacy Surveys Tractor catalog:

```
sedphot spherox --name "NGC 4889" --out-dir NGC_4889
```

If the shape lookup fails or returns nothing usable, the command **stops** rather than guessing. Three ways forward:

Give the shape yourself, as n , axis ratio (a/b , at least 1), position angle in degrees east of north, and effective radius in arcseconds:

```
sedphot spherox --name "NGC 4889" --out-dir NGC_4889 \
  --sersic-params 4 1.3 65 12
```

Fit the shape on an image instead:

```
sedphot spherox --name "NGC 4889" --sersic-from z --out-dir NGC_4889
```

Or treat the target as a point source (this biases an extended source, in a wavelength-dependent way):

```
sedphot spherox --name "NGC 4889" --model psf --out-dir NGC_4889
```

Each distinct configuration gets its own tagged output file, so a PSF run and a Sérsic run coexist. Nothing is ever overwritten.

6.15 Process a list of galaxies one after another

Put one galaxy name per line in a plain text file, `galaxies.txt`:

```
NGC 4889
NGC 4874
M87
```

Windows — Command Prompt or Anaconda Prompt batch file. Save as `run_all.bat` next to `galaxies.txt`, then double-click it or type `run_all.bat`:

```
@echo off
for /f "usebackq delims=" %%G in ("galaxies.txt") do (
  echo === %%G ===
  sedphot run --name "%%G" --out-dir "galaxies\%%G"
)
```

Windows — PowerShell. Type this at a PowerShell prompt, or save it as `run_all.ps1`:

```
Get-Content galaxies.txt | ForEach-Object {
  Write-Host "=== $_ ==="
  sedphot run --name "$_" --out-dir "galaxies\$($_ -replace ' ','_')"
}
```

macOS / Linux — bash or zsh. Save as `run_all.sh`, make it executable with `chmod`

+x run_all.sh, then run ./run_all.sh:

```
#!/usr/bin/env bash
while IFS= read -r name; do
  [ -z "$name" ] && continue
  echo "=== $name ==="
  sedphot run --name "$name" --out-dir "galaxies/${name// /_}"
done < galaxies.txt
```

Careful

Run these one galaxy at a time, not in parallel. If you are also using `--registry-update` (see section 11), running in parallel will corrupt the shared file.

7 Every flag, explained

7.1 Shared by `resolve`, `catalogs`, `measure`, `spherex`, `run`

Flag	Takes	What it does	Default
<code>--name</code>	text	Resolvable object name; tried against Sesame, then NED, then SIMBAD. Use instead of <code>--ra/--dec</code> .	none
<code>--ra</code>	number	Right ascension in decimal degrees. Requires <code>--dec</code> .	none
<code>--dec</code>	number	Declination in decimal degrees. Requires <code>--ra</code> .	none
<code>--out-dir</code>	path	Directory everything lands under. Required on every verb except <code>resolve</code> and <code>remeasure</code> .	none
<code>--label</code>	text	Output filename stem.	sanitized name, or an IAU designation for a bare position

Give exactly one of `--name` or the `--ra/--dec` pair.

7.2 `catalogs`

Flag	Takes	What it does	Default
<code>--instruments</code>	one or more names	Archives to query: <code>allwise</code> , <code>galex</code> , <code>hst</code> , <code>jplus</code> , <code>legacy</code> , <code>panstarrs</code> , <code>sdss</code> .	none
<code>--all</code>	(switch)	Query every archive. Use this or <code>--instruments</code> .	off
<code>--radius</code>	arcsec	Starting cone-search radius. Doubles up to five times if nothing is found.	2.0
<code>--legacy-dr</code>	<code>dr9</code> or <code>dr10</code>	Legacy Surveys data release. <code>dr10</code> adds i-band in the south.	<code>dr9</code>

Flag	Takes	What it does	Default
<code>--dered</code>	(switch)	Correct fluxes for Milky Way extinction. Off means as-measured, with the correction factor recorded per row.	off

7.3 measure

Flag	Takes	What it does	Default
<code>--instruments</code>	one or more names	Image archives: <code>cfht</code> , <code>hst</code> , <code>legacy</code> , <code>panstarrs</code> , <code>sdss</code> .	none
<code>--all</code>	(switch)	Every image archive. Use this or <code>--instruments</code> .	off
<code>--mode</code>	<code>aperture</code> or <code>sersic</code>	<code>aperture</code> reports the flux inside the circular aperture. <code>sersic</code> reports the fitted galaxy model's flux instead.	<code>aperture</code>
<code>--bands</code>	one or more filters	Filter subset applied to every archive.	each archive's own default set
<code>--aperture</code>	arcsec	Aperture radius. Must lie inside the <code>--radii</code> grid.	12.0
<code>--radii</code>	one or more arcsec	Radii at which the curve of growth is stored.	2 to 40 in 1-arcsec steps
<code>--cutout-size</code>	arcsec	Width of the image cutout. Half of this is the largest usable radius.	120
<code>--sky-rmin</code>	arcsec	Boundary between target and sky. Sky is measured only outside it. Lower it for compact targets.	15
<code>--registry</code>	path	JSON file of shared neighbor shapes to reuse (section 11).	none
<code>--registry-update</code>	(switch)	Also write this galaxy's results back into <code>--registry</code> . Requires <code>--registry</code> .	off
<code>--sersic-from</code>	band	With <code>--mode sersic</code> : fit the galaxy's shape on this band (<code>z</code> or <code>Legacy_z</code>) and use it for all bands.	per-archive refit
<code>--sersic-params</code>	4 numbers	With <code>--mode sersic</code> : state the shape explicitly as <code>n</code> , axis ratio (<code>a/b >= 1</code>), position angle (deg E of N), effective radius (arcsec).	none
<code>--sersic-seeing</code>	arcsec	PSF full width at half maximum assumed by a <code>--sersic-from</code> fit. The fitted index and radius are sensitive to it.	a typical value for that survey, not a measured one; the run prints a warning saying so
<code>--legacy-dr</code>	<code>dr9</code> or <code>dr10</code>	Legacy Surveys release for both images and the scene catalog.	dr9

Flag	Takes	What it does	Default
<code>--legacy-bricks</code>	(switch)	Download full Legacy brick coadds instead of small cutouts. Adds real per-pixel errors; roughly 40 MB per file.	off
<code>--hst-proposal-id</code>	text	Restrict HST to one observing program.	none
<code>--dump-arrays</code>	(switch)	Write raw per-band arrays under <code><Instrument>/QA/</code> for debugging.	off

`--sersic-from`, `--sersic-params` and `--sersic-seeing` are refused under `--mode aperture`; the command tells you so rather than ignoring them.

7.4 spherex

Flag	Takes	What it does	Default
<code>--model</code>	psf or sersic	Source model for the forced photometry. <code>psf</code> biases extended sources.	sersic
<code>--sersic-params</code>	4 numbers	Explicit shape: n (≤ 6), axis ratio ($a/b \geq 1$), position angle (deg E of N), effective radius (arcsec).	Tractor catalog lookup
<code>--sersic-from</code>	band	Fit the shape on this band's image instead of the catalog lookup.	none
<code>--sersic-seeing</code>	arcsec	PSF full width at half maximum of the shape-fit band.	a typical value for that survey, not a measured one
<code>--bkg-size</code>	pixels	Size of the tool's background-estimation region.	15
<code>--mjd-range</code>	2 numbers	Restrict to visits inside this MJD window. This is IRSA's documented workaround for epochs whose metadata breaks the job.	none
<code>--poll</code>	seconds	How often to ask IRSA whether the job is done.	5
<code>--timeout</code>	seconds	Give up after this long.	3600
<code>--legacy-dr</code>	dr9 or dr10	Legacy release for the shape lookup or shape-fit image.	dr9

7.5 sed

Flag	Takes	What it does	Default
<code>--out-dir</code>	path	Directory holding the tables. Required.	none

Flag	Takes	What it does	Default
<code>--label</code>	text	Which table family to plot.	inferred when the directory holds exactly one

7.6 overlay

Flag	Takes	What it does	Default
<code>--out-dir</code>	path	Directory holding the tables. Required.	none
<code>--label</code>	text	Which table family to plot.	inferred when unambiguous
<code>--zoom-size</code>	arcsec	Half-width of the zoom panel.	5
<code>--context-size</code>	arcsec	Half-width of the context panel.	15
<code>--wcs-from</code>	path	A local FITS file already on the composite's grid, used instead of downloading the ~380 MB detection mosaic. Refuses if the path names no file, or if the grid does not match.	none
<code>--dpi</code>	number	Figure resolution.	200

7.7 remeasure

The first argument is positional: the path to a `<label>_measured.provenance.json`.

Flag	Takes	What it does	Default
<code>--mode</code>	sersic or aperture	sersic reports the fitted model's flux; aperture reports the measured, neighbor-subtracted flux.	sersic
<code>--aperture</code>	arcsec	Radius to report at. Matches the measure default, so a bare remeasure re-reports at the radius the measurement used.	12
<code>--integrated</code>	(switch)	Report the total instead of an aperture. In sersic mode that is the model's total; in aperture mode it is the outermost stored radius. Overrides <code>--aperture</code> .	off
<code>--shape</code>	forced or fitted	Which galaxy shape the report is built on. forced uses the archive's reference-band shape. fitted uses each band's own shape where one was stored; bands without one fall back to forced and say so.	forced

Flag	Takes	What it does	Default
<code>--registry</code>	path	Registry file to fall back on when rebuilding past the stored radius grid.	a <code>registry.json</code> beside the galaxy directory, if present
<code>--write-qa</code>	(switch)	When rebuilding past the grid, write per-band figures to <code>QA/remesure_R<N>as/</code> .	off
<code>--out</code>	path	Write the result to this CSV instead of printing it.	print

7.8 run

`run` takes every flag from `catalogs` (`--radius`, `--dered`, `--legacy-dr`) and every flag from `measure` except `--instruments`, `--all` and `--dump-arrays`, plus:

Flag	Takes	What it does	Default
<code>--skip</code>	one or more names	Archives to leave out. Accepts both catalog and image archive names.	none
<code>--spherex</code>	off, psf, sersic	Also fetch SPHEREx, with this source model.	off

`run` uses every archive that is not skipped, which is why it has no `--instruments`.

One special case: `--sersic-params` is normally refused under `--mode aperture`, but under `run` with `--spherex psf` or `--spherex sersic` it is accepted, because there it also declares the shape SPHEREx extracts with. One shape per galaxy, shared by both stages.

8 What comes out

8.1 The directory tree

A full run of one galaxy produces:

```

<out-dir>/
+-- patches.json                (optional, you create it)
+-- Photometry/
    +-- <label>_catalog.csv      archive photometry
    +-- <label>_catalog.provenance.json
    +-- <label>_measured.csv     measured photometry
    +-- <label>_measured.provenance.json
    +-- <label>_sed.png          combined SED figure
    +-- <label>_overlay.png      position check figure
    +-- coverage_catalogs.json  per-archive outcome
    +-- coverage_measure.json   per-archive outcome
    +-- Legacy/                 cached images + QA/Legacy_<band>.png
    +-- SDSS/                   cached images + QA/SDSS_<band>.png
    +-- PanSTARRS/              cached images + QA/PS1_<band>.png
    +-- CFHT/                   cached images + QA/CFHT_<band>.png

```

```

+-- HST/          cached mosaics + QA/HST_<band>.png
+-- QA/
|   +-- growth_curves.png          all bands, one figure
+-- scene/          cached catalog queries
+-- SPHEREx/
    +-- table_photometry.<tag>.csv
    +-- table_photometry.<tag>.provenance.json
    +-- extractions.json

```

8.2 What each file is

File	What it is
<label>_catalog.csv	One row per band, from the published archive catalogs. The nearest source to the requested position in each archive.
<label>_measured.csv	One row per band, measured by sedphot from the images. Same column layout as the catalog table.
*.provenance.json	The complete record of how the neighboring file was made: every setting, the code revision, a checksum of the file, and for measurements the full fit. This is what remeasure reads. Do not edit it.
<label>_sed.png	Flux against wavelength, both tables together. Catalog points are filled, measured points are open. Marker shape identifies the instrument, color tracks wavelength.
<label>_overlay.png	Two panels of the HST color image with each archive's matched position marked.
coverage_catalogs.json, coverage_measure.json	Per-archive outcome: ok , no_coverage , no_match , or error , with a message and the number of rows.
<Instrument>/*.fits	The downloaded images, cached. Deleting one forces a re-download on the next run.
<Instrument>/QA/ <Inst>_<band>.png	The per-band diagnostic figure (section 9). <Inst> is the instrument's short name, which matches the directory everywhere except PanSTARRS/ , whose figures are named PS1_<band>.png .
QA/growth_curves.png	Enclosed flux against radius for every measured band on one axis.
scene/*.csv	Cached catalog queries describing the neighboring sources. Deleting these forces a re-query.
SPHEREx/table_photometry. <tag>.csv	The raw per-visit SPHEREx table, exactly as IRSA returned it. Never overwritten.
SPHEREx/extractions.json	An index mapping each tag to the configuration that produced it.

8.3 The output table columns

Both <label>_catalog.csv and <label>_measured.csv share this layout.

Column	Meaning	Units
band	Band label, <Inst>_<filter> — e.g. Legacy_g, SDSS_u, PS1_z, CFHT_r, GALEX_FUV, WISE_W1, JPLUS_uJAVA, HST_F475W.	—
flux_uJy	The flux.	microjansky (AB)
flux_err_uJy	Uncertainty on the flux.	microjansky
mag_AB	The same measurement as an AB magnitude. Empty when the flux is zero or negative.	mag
mag_err	Uncertainty on the magnitude.	mag
target_ra	The position you asked for.	degrees
target_dec	The position you asked for.	degrees
match_ra	Where the archive’s matched source actually is. For measured rows this equals <code>target_ra</code> .	degrees
match_dec	Where the archive’s matched source actually is.	degrees
sep_arcsec	Distance from the requested position to the match. 0 for measured rows.	arcsec
flags	Quality tokens, <code>key=value</code> separated by semicolons (see below). For catalog rows, whatever flags that archive publishes.	—
source	Where the number came from — Legacy_DR9, SDSS_DR17_cModel, PanSTARRS_DR1, GALEX_GUVcat_AIS, sedphot_aperture_scene_empap, and so on.	—
retrieved	Date the value was pulled.	ISO date
mw_transmission	Milky Way transmission in this band, where the archive supplies it. Empty when unknown.	fraction
dered_applied	True if the flux has been corrected for Milky Way extinction. Measured rows are always False.	—

Three things to know about the numbers.

- **Fluxes are microjansky on the AB system.** Convert to AB magnitude with $\text{mag} = 23.9 - 2.5 \log_{10}(\text{flux})$.
- **The errors are statistical only.** They describe the noise in the measurement and nothing else — no calibration uncertainty, no error floor, no model uncertainty. Adding those is the SED fitter’s job, not this table’s.
- **Negative fluxes are real.** A non-detection in a faint band gives a small negative number after background subtraction. That is the honest measurement and it is preserved deliberately; it is not an error, and it must not be clipped to zero. Its `mag_AB` cell is simply empty.

The flags tokens on measured rows. These let you filter a table without opening a single figure.

Token	Meaning	What to look for
cov	Fraction of aperture pixels with real data.	Should be 1.000. Below 0.95 the band is refused outright.

Token	Meaning	What to look for
maskfrac	Fraction of the aperture masked because a neighboring source sat there.	A few percent is normal. Above ~0.2 the flux leans heavily on reconstruction.
twinfrac	Fraction of masked pixels reconstructed from the galaxy's mirror side rather than from the model.	1.00 is best.
nbsub	Flux subtracted from the aperture as neighboring sources.	Compare against the reported flux. Large values mean a crowded field.
excess	Flux growing outside the aperture beyond what the galaxy's own model predicts.	Near zero is good. Large positive means contamination or an unmodeled envelope.
pedb	Residual uniform background left in the curve of growth.	Should be near zero.
conv	Radius, in arcsec, at which the curve of growth flattened. 0 means it never did.	Should be at or inside the aperture.
bg	The fitted background level.	—
far	An independent background level measured far from the target.	Should broadly agree with bg .
mesh	Flux the smooth background surface removed from the aperture.	Large values mean structured background.
art	Area, in square arcsec, flagged as a detector artifact (a bleed trail, a streak).	Present only when one was found.
leash	How many neighboring sources hit their brightness limit.	Present only when some did.
refit	The galaxy's fitted brightness relative to the catalog's.	Near 1 is unremarkable.
atbound	How many fitted shape parameters ended at a limit.	Present only when some did.
reg	How many neighbors came from the shared registry.	—
scene=none	No catalog of neighboring sources was available; the galaxy was measured with no neighbor modeling at all.	Treat the flux with care in a crowded field.

9 Reading the QA figures

Two figures per measurement run, plus one per band.

9.1 The per-band figure

`<Instrument>/QA/<Inst>_<band>.png` — five panels in a row. The prefix is the instrument's short name, which matches the directory everywhere except PanSTARRS/, whose figures are `PS1_<band>.png`.

1. **data** — the image, with stars already subtracted. The blue circle is your aperture. Orange contours, if present, mark detector artifacts. The title gives the fitted background level and its tilt.
2. **fitted scene + background** — the model: the galaxy, its neighbors, and the background, as the fit believes them to be.
3. **residual - mesh** — the data minus the model, and minus the smooth background surface. This is the panel that tells you whether the fit worked.
4. **masked + filled - mesh** — the image the flux is actually measured from: masked pixels replaced by reconstructed values. Green contours outline the mask. The title gives the masked fraction of the aperture.
5. **curve of growth** — enclosed flux against radius. The red line with dots is the measurement; the black dashed line is the fitted galaxy model. The blue vertical line is your aperture; a grey dotted line marks where the curve flattened. The title gives the final flux, its error, and the excess growth.

A good figure.

- Panel 3 is flat noise. No bright leftover at the galaxy's center, no dark hole where a neighbor was over-subtracted.
- Panel 5's red curve rises and then flattens, and the flattening happens at or inside the blue aperture line. The title says `conv@14"` or similar rather than `no plateau`.
- The red curve and the black dashed model track each other out to the aperture.
- `excess` in the title is small compared with the flux.
- Green mask contours in panel 4 cover only a small part of the blue circle.

A suspicious figure.

What you see	What it means
Title says <code>no plateau</code>	The curve of growth was still rising at 40 arcsec. Either the galaxy is genuinely large, or the background is under-subtracted. Check <code>pedb</code> and the <code>far</code> token.
Red curve turns over and falls	The background was over-subtracted.
Large positive <code>excess</code>	Light in the outskirts that the galaxy's model does not account for — a neighbor, or a real envelope.
Dark hole at a neighbor's position in panel 3	That neighbor was over-subtracted.
Bright residual core in panel 3	The galaxy profile did not fit; the flux may be biased.
Green contours across much of the blue circle	Most of the aperture is reconstructed rather than measured. Check <code>twinfrac</code> .
Orange artifact contours crossing the aperture	A detector artifact sits on the target.
Red and black curves diverge inside the aperture	The fitted model is a poor description of the galaxy. Fine in <code>--mode aperture</code> (the model is only a cross-check there), serious in <code>--mode sersic</code> (the model <i>is</i> the answer).

9.2 The growth-curve figure

QA/growth_curves.png — every measured band on one axis, enclosed flux against radius, with a logarithmic vertical scale. Line style identifies the instrument, color tracks wavelength. The blue vertical line is the science aperture.

Read it for consistency across bands. Curves should be roughly parallel and should flatten at similar radii. A single band that keeps climbing while its neighbors flatten, or one that crosses the others, is the band to open individually.

10 When something goes wrong

Every command reports failures rather than hiding them. Most messages name the fix. The recurring ones:

Symptom	Cause	Fix
error: the following arguments are required: --out-dir	--out-dir is required on every verb that writes.	Add --out-dir <directory>.
error: give --instruments ... or --all	catalogs and measure need to be told which archives to use.	Add --all, or --instruments legacy sdss.
Could not resolve 'X' with Sesame, NED, or SIMBAD	The name is not in any resolver.	Look the position up yourself and use --ra / --dec.
--registry-update needs --registry PATH	You asked to write to a registry without naming one.	Add --registry <path>, or drop --registry-update.
--sersic-from only applies to --mode sersic	A shape flag was given under aperture mode, where it would have no effect.	Add --mode sersic, or drop the flag.
ValueError: --aperture 60" lies outside the curve-of-growth grid [2, 40]"	The aperture is outside the radius grid the run stores.	Pass --radii covering it, and a --cutout-size at least twice the largest radius.
An image archive reports no_coverage	That survey's footprint does not include this position. Nothing is wrong.	Nothing to do. Fewer bands is the correct answer.

Symptom	Cause	Fix
A catalog archive reports no_match	Either the field genuinely holds no catalogued source within the expanded search radius (including “this position is outside that survey’s footprint”), or the archive was down. The two are indistinguishable: a failed query returns no rows, and the radius expansion then reports no_match . Catalog archives never report no_coverage .	Re-run just that archive later. If it still says no_match , try a larger --radius . An unexpected no_match on a well-covered galaxy is almost always an outage.
An archive reports error	A service or parsing failure that got past the provider’s own retries. The message names it.	Read coverage_catalogs.json or coverage_measure.json , then re-run that archive.
Legacy z: no_coverage --aperture coverage 0.83 < 0.95	More than 5% of the aperture is off the image footprint, blank, or covered by an artifact. The band is refused rather than reporting a silently biased flux.	Use a smaller --aperture , or a larger --cutout-size if the cutout was simply too small. Otherwise accept the missing band.
blank pixels inside the 0.5" peak	Dead pixels at the very center of the galaxy. No reconstruction can invent the peak.	The band cannot be measured.
scene catalog is empty here; measuring blind, and scene=none in flags	No catalog of neighboring sources covers this position, so nothing was subtracted or masked.	The flux is still measured, but with no neighbor handling. In an uncrowded field this is fine; in a crowded one, treat it with care.
No images fetched from any provider; nothing to measure.	No archive covered the position, or all of them failed.	Check coverage_measure.json for which and why.
overlay downloads ~380 MB	The HST color composite carries no coordinate grid of its own, so the detection mosaic is fetched for its grid. It is cached after the first time.	Pass --wcs-from with a local FITS file already on that grid.

Symptom	Cause	Fix
no HAP total/detection product at this position, or sedphot overlay: no figure written --wcs-from names no file	HST has not imaged this field. The path is mistyped. It refuses rather than falling back to the big download.	The overlay is unavailable for this galaxy. Correct the path.
[spherex] Tractor shape lookup failed ... aborting or no usable extended Tractor shape at this position table_photometry. <tag>.csv exists but no sidecar vouches for this configuration [spherex] WARNING: reff 0.60" is below the tool's 1" point-source threshold remeasure prints aperture 45" is past the stored curve of growth (ends at 40"); rebuilding the scene sedphot remeasure: aperture 70" exceeds the stamp half-width (60") remeasure says bands fell back to the forced shape under --shape fitted	SPHEREx will not guess a source shape. A raw table written under the wrong shape is permanent and unmarked, so it stops instead. A raw SPHEREx table is on disk that nothing certifies. Nothing is ever overwritten. The tool treats anything that small as a point source, so the Sérsic parameters do nothing. Beyond the stored grid there is no value to interpolate, so the scene is rebuilt and re-integrated. The requested radius runs off the cutout the fit was built on. Those bands have no per-band shape stored — either an older measurement, or a galaxy where the extra fit never ran.	Pass --sersic-params N AXR PA RE, or --sersic-from z to fit one, or --model psf. Move the file aside deliberately, then re-run. Either accept it, or use --model psf and be explicit. Nothing wrong — just expect minutes rather than milliseconds. Re-measure with a larger --cutout-size. Either accept forced, or re-measure the galaxy.

Symptom	Cause	Fix
cannot infer --label in ...: found stems [...]	The directory holds tables for more than one galaxy, so <code>sed</code> and <code>overlay</code> cannot guess which.	Pass <code>--label <stem></code> .
[sed] WARNING ... the plotted fluxes are NOT on a common extinction scale	You used <code>--dered</code> , which corrects the catalog table only. The measured table is always as-measured, so the figure mixes the two.	Expected. Do not read colors off that figure; use the tables.

11 Getting more out of it

Two features exist for cases the defaults do not cover. You will not need either at first.

11.1 `patches.json` — correcting the scene by hand

Before measuring, `sedphot` builds a picture of the field from the Legacy Surveys catalog: which sources are there, how big and how bright. Occasionally that catalog is wrong in a way that matters — most often it merges a close pair into a single entry, or splits one galaxy into several.

You can correct it. Put a file named `patches.json` in the galaxy's `--out-dir` (next to `Photometry/`, not inside it). It is read automatically and its keys are logged at the start of the run.

The keys it accepts:

Key	What it does
<code>replace_rows</code>	Replace one catalog entry with one or more of your own. The usual fix for a merged close pair.
<code>free_seats</code>	Let the source at a given position have its shape fitted rather than taken from the catalog.
<code>target_system</code>	Declare that a nearby source belongs to the target — its light is counted with the galaxy rather than subtracted from it.
<code>target_halo</code>	Give the target an extended-envelope component, for a brightest-cluster galaxy.
<code>target_refit</code>	Turn the target's own shape refit on or off.
<code>snap_gated</code>	Move fitted sources onto their nearest brightness peak.
<code>harvest_target</code>	Record the target's own shape into the shared registry.
<code>comment</code>	Free text. Ignored by the code; use it to say why the patch exists.

Reach for this when a QA figure shows a clear over- or under-subtraction at a specific neighbor, and only then. An unnecessary patch is a way to bias a measurement.

11.2 The cross-field registry — shared neighbors

When several galaxies in the same field are measured separately, a neighboring source may appear in more than one of them. Left alone, each measurement fits that source independently, and the fits disagree.

The registry fixes that. It is one JSON file recording the fitted shape, position and per-band brightness of shared sources, in sky units so an entry works from any field.

Consume it:

```
sedphot measure --name "M87" --all --out-dir M87 \  
  --registry registry.json
```

Consume it and write this galaxy's results back into it:

```
sedphot measure --name "M87" --all --out-dir M87 \  
  --registry registry.json --registry-update
```

Two things to know before using it:

- **Run galaxies one at a time.** Updates are last-writer-wins with no file locking. Two runs writing at once will corrupt the file.
- **Order matters while the file fills.** A galaxy measured after its neighbor was registered gets that neighbor's shape frozen in rather than fitted fresh. Once every galaxy in a field has been measured once, a second pass over the same file is uniform.

Use it when you are measuring several galaxies in one field and want their shared neighbors treated identically. Skip it for a single galaxy; it buys nothing there.

Note

Every number this package writes is accompanied by a `.provenance.json` file recording the settings, the code revision, and a checksum of the product. When a result needs defending, that file is the record.